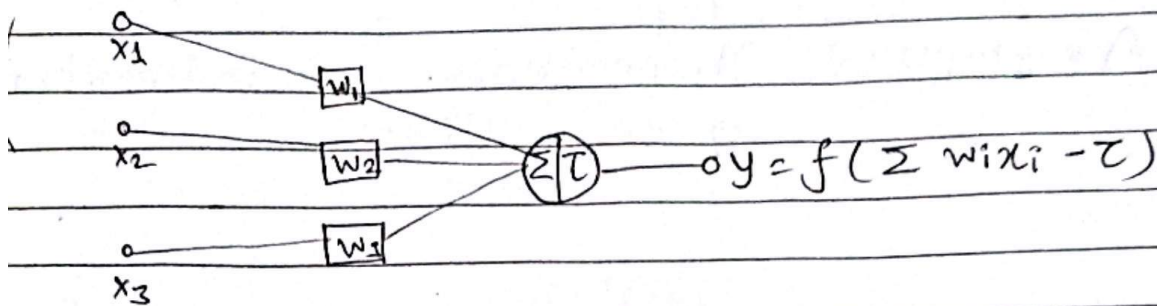


Assignment 4

Task 1:

To understand how a neuron can be used to store information



If $x_1 = 1, x_2 = 1, x_3 = 1$ and $w_1 = 1, w_2 = 2$ and $w_3 = 1$, $\tau = 2$,

What would the value of y be?

Use

$$y = \sum w * x - \tau \text{ which is } y = (1*1 + 1*1 + 1*1 - 2) = 1$$

Q1) If the weights are the same but $x_1 = 0, x_2 = 0, x_3 = 0$, what is the value of y ?

Ans:

$$y = 1 \times 0 + 1 \times 0 + 1 \times 0 - 2 = -2 \neq$$

Q.2) If the weights are the same but $x_1 = 1, x_2 = 0, x_3 = 0$, what is the value of y ?

Ans:

$$y = 1 \times 1 + 1 \times 0 + 1 \times 0 - 2 = -1 \neq$$

Q.3) If the weights are the same but $x_1 = 0, x_2 = 1, x_3 = 0$, what is the value of y ?

Ans:

$$y = 1 \times 0 + 1 \times 1 + 1 \times 0 - 2 = -1 \neq$$

Q.4) If the weights are the same but $x_1 = 0, x_2 = 1, x_3 = 1$, what is the value of y ?

Ans:

$$y = 1 \times 0 + 1 \times 1 + 1 \times 1 - 2 = 0 \neq$$

Q. What would happen if add at the end of this neuron a further function is added that says if $y \geq 0$ then output is 1 else output is 0.

Ans:

Here,

$$f(y) = \begin{cases} 1, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

So,

$$\text{for } y = -2, \rightarrow f(y) = 0$$

$$\text{for } y = -1, \rightarrow f(y) = 0$$

$$\text{for } y = -1, \rightarrow f(y) = 0$$

$$\text{for } y = -1 \rightarrow f(y) = 0$$

$$\text{for } y = 0 \rightarrow f(y) = 1$$

$$\text{for } y = 1 \rightarrow f(y) = 1$$

Q. Using this new function and the neuron, what parameters would be needed to produce an OR gate.

Ans: If we use a 3-input gate,

$$w_1 = w_2 = w_3 = 1$$

$$\text{and } T = 1$$

$$\text{Then, } y = x_1 + x_2 + x_3 - 1$$

After the threshold function;

Inputs	Sum	y	Output
000	0	-1	0
100	1	0	1
010	1	0	1
001	1	0	1
110	2	1	1
101	2	1	1
011	2	1	1
111	3	2	1

Task 2

Ans: The given neural network has 2 inputs given by sensors and 2 neurons (outputs).

Each neuron computes

$$y = f(w_1x_1 + w_2x_2 + b)$$

Output:

$$\text{Output} = \begin{cases} 1, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

Optim 1 for Output 1

For output 1

Consider $b = -0.5$

$$w_{11} = 1$$

$$w_{21} = 0$$

$$\text{So, } y = -0.5 + 1x_1 + 0x_2$$

x_1	x_2	Calculation	y	Output
0	0	$-0.5 + 1 \times 0 + 0 \times 0$	-0.5	0
0	1	$-0.5 + 1 \times 0 + 0 \times 1$	-0.5	0
1	0	$-0.5 + 1 \times 1 + 0 \times 0$	0.5	1
1	1	$-0.5 + 1 \times 1 + 0 \times 1$	0.5	1

This matches the required output 1.

$\therefore$ Therefore, weights = $\{-0.5, 1, 0\}$ or $\{-0.5, 0, 1\}$

Output 2:

Consider

$$b = -0.5$$

$$w_{12} = 1$$

$$w_{22} = 1$$

$$\text{So, } y = -0.5 + 1x_1 + 1x_2$$

x_1	x_2	Calculation	y	Output
0	0	$-0.5 + 0 \times 0 + 1 \times 0$	-0.5	0
0	1	$-0.5 + 0 \times 0 + 1 \times 1$	0.5	1
1	0	$-0.5 + 0 \times 1 + 1 \times 0$	-0.5	0
1	1	$-0.5 + 0 \times 1 + 1 \times 1$	0.5	1

Therefore,

$$\text{weights} = \{-0.5, 1, 0\} \text{ or } \{-0.5, 0, 1\}$$